Web Service - Assignment 1(10%)

Instructions:
This assignment must be completed individually. Group submissions are not allowed. Each student is required to design, implement, test, and document their own **ShapeTool** SOAP Web Service project independently. Any similarity in code, XSD structure, or documentation that indicates copying will result in academic penalties. All students must submit the complete project package, including the XSD file, WSDL file, source code, sample SOAP request and response messages, and documentation. The submission must be organized clearly in a GitHub repository then submit the **GitHub** link:
In addition to submission, each student is required to present the project to the instructor on **April 8, 2026**. During the presentation, students must:

- Explain the XSD design and schema structure
- Describe the WSDL and service operations
- Demonstrate the running SOAP web service
- Show live testing using SoapUI or Postman
- Explain how validation and SOAP Fault handling were implemented

The presentation should take approximately **6–8 minutes per student**, followed by questions from the instructor.

Assignment: Developing a SOAP Web Service – *ShapeTool*

Design and implement a SOAP-based web service called **ShapeTool** whose purpose is to calculate the area of different geometric shapes. The main objective of this project is to understand the principles of Service-Oriented Architecture (SOA), SOAP messaging structure, XML Schema (XSD) design, WSDL definition, and contract-first web service development. You are required to follow a contract-first approach, meaning you must first design the XML Schema (XSD) that defines the structure of the request and response messages before implementing the actual web service logic. The **ShapeTool** web service must expose separate operations (methods) for each shape. The required operations are:

- circleArea
- squareArea
- rectangleArea
- parallelogramArea
- triangleArea

Each operation must accept only the parameters required for that specific shape. For example, circleArea must accept a radius value, rectangleArea must accept length and width, triangleArea must accept base and height, and so on. The service should compute the area using the correct mathematical formula and return the result in a structured SOAP response message specific to that operation.

The XML structure for each request and response must be clearly defined in the XSD file. You must define proper complex types, use appropriate XML data types (such as xs:double), apply restrictions to prevent invalid input values (such as negative numbers), and properly define namespaces. The web service must use document/literal style messaging to comply with SOAP standards.

After designing the XSD, you must generate or create the corresponding WSDL file that defines all service operations (circleArea, squareArea, rectangleArea, parallelogramArea, and triangleArea), their input and output messages, binding configuration, and service endpoint. You will then implement the SOAP web service using a suitable framework such as Java (JAX-WS or Spring Boot). The implementation must validate incoming requests against the XSD and handle possible errors such as missing parameters, invalid values, or negative numbers by returning appropriate SOAP Fault messages.

Finally, you must test your web service using tools such as **SoapUI or Postman** and provide sample SOAP request and response messages for each operation as evidence of proper functionality. The project submission must also include brief documentation explaining your design decisions, schema structure, namespace usage, validation strategy, service operation design, and error handling mechanisms.